Muhammad Talal Akhtar

Computer Science Student | Aspiring Software Engineer

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GitHub Profile

SUMMARY

Computer Science student at FAST-NUCES with a strong interest in full-stack development. Experienced in building MERN stack applications, including designing RESTful APIs, implementing JWT-based authentication, and working with efficient data models.

Possesses a solid foundation in data structures and algorithms, with hands-on experience through C++ projects. Enjoys writing clean, modular code and building practical solutions to real-world problems, with a focus on developing reliable and scalable applications.

TECHNICAL SKILLS

Languages: C++ (Advanced), JavaScript, Python
Frontend: React.js, HTML5, CSS3, Responsive Design
Backend: Node.js, JWT Authentication, bcrypt
Databases: MongoDB + Mongoose
Core Concepts: Data Structures & Algorithms, OOP, Graph Theory, Stack/Queue
Tools: Git, GitHub, VS Code
Practices: Debugging, Performance Optimization, Code Reviews

PROJECTS

Personal Finance Tracker **GitHub ↗**

Node.js | Express.js | React.js | MongoDB | JWT | REST APIs

- Architected a MERN application with secure JWT + bcrypt authentication, ensuring protected access to all users.
- Developed 12+ RESTful API endpoints for transaction management with structured JSON responses.
- Designed MongoDB schemas with Mongoose validation, enforcing data integrity and reducing invalid writes.
- Optimized API communication to reduce redundant requests and improve application responsiveness.

Advanced Social Network System **GitHub ↗**

C++ | Data Structures | Graph Algorithms | OOP

- Designed a terminal-based backend system modeling user relationships using weighted graphs.
- Implemented BFS and DFS algorithms for relationship traversal and friendship discovery features.
- Applied modular OOP principles with decoupled classes, improving scalability and code maintainability.
- Demonstrated strong understanding of graph theory and system-level problem solving in C++.

Personal Portfolio Website **Live ↗**

HTML5 | CSS3 | JavaScript | Responsive Design

- Developed a fully responsive, mobile-first portfolio website with optimized performance.
- Achieved fast load times through lightweight design and efficient asset handling.
- Implemented interactive UI elements and smooth navigation using vanilla JavaScript.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad
Bachelor of Science in Computer Science